

ORIGINAL ARTICLE

FuzzyGNN: a graph neural network framework for enhanced fuzzy modeling

Gianluca Zaza · Raffaele Scaringi · Gennaro Vessio · Giovanna Castellano

Received: 10 September 2025 / Accepted: 3 February 2026

© The Author(s) 2026

Abstract

Balancing predictive performance with transparency remains a core challenge in eXplainable Artificial Intelligence, especially for tabular data. In this work, we present FuzzyGNN, a novel hybrid framework that combines fuzzy logic with Graph Neural Networks (GNNs) to deliver both high accuracy and interpretability. FuzzyGNN represents each data instance as a graph, where raw features and fuzzy sets are modeled as nodes, enabling contextual reasoning through message passing. Unlike traditional neuro-fuzzy systems—often limited by scalability and rule explosion—FuzzyGNN enforces model compactness through its graph-aware architecture and a pruning mechanism guided by graphbased explanations. Extensive experiments on seven benchmark datasets show that FuzzyGNN achieves competitive predictive performance while drastically reducing the number of fuzzy rules, linguistic terms, and active features used to produce output fuzzy rules. Additionally, it offers structural interpretability through rule extraction and post-hoc explainability via GNNExplainer. These capabilities make FuzzyGNN particularly well-suited for domains where transparency is as crucial as accuracy, such as healthcare, finance, and safety-critical systems. The code is available in the following link: <https://github.com/cilabuniba/fuzzygnn>

Keywords Fuzzy inference systems · Graph neural networks · Explainable artificial intelligence · Model interpretability · Neuro-fuzzy systems

1 Introduction

Artificial Intelligence (AI) has made remarkable progress in recent years, achieving impressive performance across a wide range of tasks. However, this success has often come at the cost of transparency. Many high-performing models—especially those based on deep learning—are regarded as “black boxes” due to their complex, opaque inner workings [1]. As AI systems are increasingly deployed in sensitive, high-stakes domains such as healthcare, finance, and public policy, the need for models that not only perform well but also provide insights into their decision-making processes has become a critical priority.

This demand has led to the emergence of eXplainable Artificial Intelligence (XAI), a research field that aims to make AI systems more understandable to humans [2]. In this context, two complementary goals have emerged: *interpretability*, which refers to the degree to which a human can comprehend a model’s internal structure, and



explainability, which concerns the ability to provide meaningful justifications for specific outputs. Striking a balance between these goals and predictive performance remains a significant challenge in designing trustworthy AI systems.

Among the most promising approaches to this challenge are Fuzzy Inference Systems (FISs), which emulate human reasoning through the use of linguistic variables and rule-based logic [3]. FISs are inherently interpretable, making them attractive for applications where transparency is essential. However, traditional fuzzy systems struggle with scalability: as the number of input features increases, the number of rules grows exponentially, leading to reduced comprehensibility and increased computational costs [4]. Furthermore, these systems often rely on domain experts to define fuzzy rules and sets, which introduces subjectivity and limits generalization [5].

To address these limitations, we propose FuzzyGNN, a novel framework that integrates fuzzy logic with Graph Neural Networks (GNNs) [6] to build models that are both expressive and transparent. Our method transforms tabular data into graph-structured representations, capturing relationships between raw input features and their fuzzified counterparts—each associated with human-readable linguistic terms. Unlike conventional neuro-fuzzy systems that use dense feedforward architectures, FuzzyGNN leverages the relational inductive bias of GNNs to model the interaction between features and fuzzy sets more effectively.

On the other hand, while GNNs excel at modeling relational data, they lack inherent interpretability. Node embeddings and message passing are often opaque, and post-hoc methods like GNNExplainer [7] provide only local, low-level insights without symbolic structure. FuzzyGNN addresses these limitations by integrating fuzzy logic into the GNN architecture: linguistic labels enhance feature semantics, while the graph structure captures contextual dependencies. This hybrid design merges the scalability of GNNs with the interpretability of FISs. Our framework builds on initial attempts to integrate fuzzy logic with GNNs [8] and extends them by introducing an automated mechanism for learning fuzzy sets, thereby removing the need for manual definition and enabling broader applicability.

One of the key innovations of FuzzyGNN is its ability to produce interpretable, instance-level explanations. By using GNNExplainer, we extract the most influential subgraphs responsible for a given prediction and translate them into concise IF-THEN rules. This approach bridges the gap between symbolic reasoning and data-driven learning, enabling factual, human-friendly justifications. In addition, we introduce a fuzzy set pruning mechanism that iteratively removes redundant or low-importance fuzzy terms, simplifying the model while preserving accuracy.

We validated our framework through extensive experiments on a diverse collection of tabular datasets, comparing FuzzyGNN against several state-of-the-art models, including classical interpretable methods (e.g., Decision Trees, Logistic Regression), black-box learners (e.g., SVM, MLP), and neuro-fuzzy systems (e.g., ANFIS). Our results show that FuzzyGNN achieves competitive accuracy while significantly improving transparency, scalability, and interpretability. In particular, it overcomes the curse of dimensionality that often hampers traditional fuzzy systems, offering a flexible and efficient alternative.

In summary, FuzzyGNN combines the structural transparency of fuzzy systems with the expressive power of graph-based deep learning, resulting in a robust yet interpretable AI solution. Its ability to provide understandable, rule-based explanations makes it particularly suitable for applications requiring both high performance and human trust.

The rest of the paper is structured as follows. Section 2 discusses related work. Section 3 introduces our proposed methodology. Section 4 reports on experimental results and evaluation. Section 5 concludes the paper and outlines directions for future research.

2 Related work

Deep learning models have achieved state-of-the-art performance across a wide range of domains. However, their lack of transparency remains a significant concern, particularly in critical applications where understanding model decisions is essential. To mitigate this issue, several post-hoc explainability methods have been developed. Among the most widely used are Local Interpretable Model-agnostic Explanations (LIME) [9] and SHapley Additive exPlanations (SHAP) [10]. LIME generates local surrogate models that approximate the decision boundary of complex models using simpler interpretable ones, while SHAP assigns additive importance values to each feature to quantify their contribution to individual predictions. Despite their utility, these methods present notable limitations: LIME may produce inconsistent explanations for similar inputs due to its reliance on random perturbations [11], and SHAP, while theoretically grounded, can be computationally expensive and susceptible to adversarial manipulation [12]. In image analysis, saliency maps are often used in conjunction with Convolutional Neural Networks to highlight regions that influence predictions. However, these visual explanations can be unstable and vulnerable to adversarial noise, undermining their reliability [13]. Overall, these techniques often provide shallow or fragile insights that fail to reveal the deeper reasoning processes of complex models.

In parallel, Graph Neural Networks have emerged as a powerful paradigm for learning over structured and relational data. Their ability to capture dependencies and contextual relationships among entities has made them valuable in tasks ranging from molecular analysis to social network modeling [14]. Techniques such as GNNExplainer [7], XGNN [15], and PGExplainer [16] have been introduced to improve the interpretability of GNNs by identifying subgraphs and features most influential to a given prediction. However, the application of GNN-based explainability techniques to traditional structured data remains underexplored, and their ability to produce human-readable justifications is still limited.

Fuzzy logic represents another well-established framework for achieving interpretability in AI systems, as it models human reasoning through linguistic variables and rule-based inference [17]. The recent interest in XAI, amplified by policy frameworks such as the GDPR's "right to explanation" and initiatives such as DARPA's interpretable AI programs [18], has revitalized research on fuzzy-based approaches. Nevertheless, traditional Fuzzy Inference Systems suffer from scalability issues: as the number of input features grows, the combinatorial explosion of fuzzy rules leads to reduced comprehensibility and higher computational burden [4]. Moreover, these systems typically depend on expert-crafted rules and membership functions, introducing bias and limiting generalizability [5].

To mitigate these challenges, hybrid neuro-fuzzy models have been proposed, integrating neural networks with fuzzy logic to automate rule learning and enhance scalability [19–22]. For example, the Fuzzy Restricted Boltzmann Machine (FRBM) introduces fuzzy-valued parameters to improve robustness [23], while the Fuzzy Deep Neural Network (FDNN) combines fuzzy encoding with deep architectures. However, in both cases, the opacity of the underlying neural components compromises the interpretability gains, rendering them closer to traditional black-box models.

Other approaches, such as the Takagi–Sugeno Deep Fuzzy Network (TSDFN) and the Fuzzy Deep Belief Network (FDBN), attempt to scale fuzzy systems for high-dimensional settings. Nevertheless, they also face the curse of dimensionality, in which increased rule complexity reduces transparency [24, 25]. Among the most widely known hybrid models, the Adaptive-Network-based Fuzzy Inference System (ANFIS) offers an interpretable architecture for learning fuzzy rules through a combination of backpropagation and least-squares estimation. However, despite its conceptual elegance, ANFIS struggles with high-dimensional features and can be computationally intensive when applied to large datasets [26].

More recent architectures, such as the Fuzzy Stacked Autoencoder (FSAE), incorporate fuzzy inference into the layers of deep learning autoencoders [27]. However, in these models, interpretability is typically confined to the final output layer, while the inner representation remains opaque. Another recent contribution is PLE-NARY [28], a multi-step framework designed to enhance the interpretability of black-box models trained on

tabular data. PLENARY combines neural learning with mid-level label supervision and post-hoc SHAP explanations, which are subsequently translated into fuzzy linguistic summaries. While the method improves the user-friendliness of explanations, it still relies on a separate explanation pipeline applied after training a black-box model, rather than embedding interpretability directly into the predictive process. In contrast, we aim to integrate fuzzy logic directly into the predictive process and enable instance-specific, structure-aware reasoning via GNNs.

In light of these limitations, we propose FuzzyGNN, a novel hybrid framework that combines fuzzy reasoning with graph-based representation learning. Specifically, FuzzyGNN transforms tabular data into graph-structured instances, where nodes correspond to raw features and their associated fuzzy sets. This structure enables the use of a GNN backbone to model contextual interactions between features and linguistic terms. By treating each data point as an individual graph, FuzzyGNN maintains a constant number of learnable parameters regardless of input size and facilitates the generation of concise, human-interpretable rules.

In doing so, FuzzyGNN addresses the key limitations of existing neuro-fuzzy systems: it scales effectively with high-dimensional data, supports factual, instance-level explanations, and avoids relying on exhaustive rule enumeration. These characteristics make it particularly well-suited for applications where explainability is not just desirable but required.

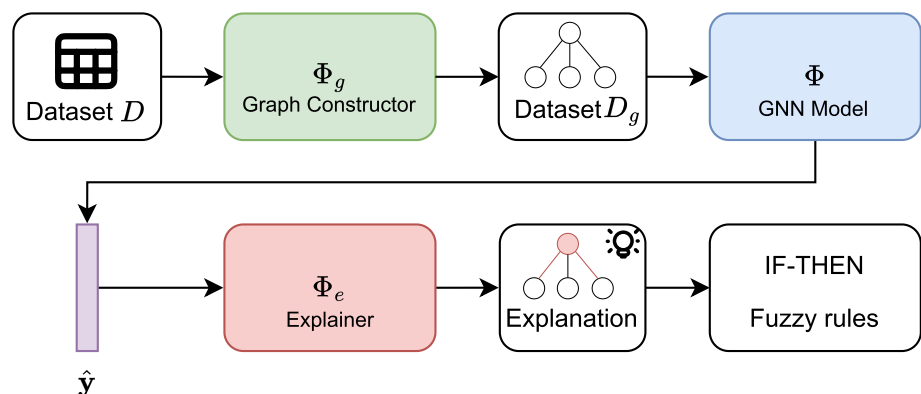
3 Method

We introduce FuzzyGNN, a novel end-to-end deep learning framework that integrates fuzzy logic with graph-based representation learning to enhance model interpretability without sacrificing predictive performance. Explicitly designed for tabular data, our approach transforms each input instance into a structured graph representation, enabling the integration of fuzzy reasoning within a Graph Neural Network architecture. As illustrated in Fig. 1, FuzzyGNN comprises three main components: the *Graph Constructor*, which converts raw input data into graph instances; the *GNN model*, which performs the classification task; and the *Explainer*, which provides factual, rule-based explanations and supports an iterative fuzzy set pruning strategy to retain only the most informative features.

3.1 Graph construction

Let $D = \{(x_d, y_d)\}_{d=1}^N$ be a tabular dataset consisting of N instances, where each data point $x_d \in \mathbb{R}^F$ is described by F numerical features, and $y_d \in C = \{C_1, \dots, C_K\}$ is the corresponding class label from one of K possible classes. The goal of the graph construction module is to convert each x_d into a graph instance g_d that incorporates both the original features and their fuzzy representations.

Fig. 1 Overview of the FuzzyGNN pipeline. The key components include: the Graph Constructor, which transforms raw tabular data into labeled graphs; the GNN Model, responsible for predicting classes based on the input graph; and the Explainer, which can either elucidate individual predictions or recommend fuzzy set pruning



To achieve this, we assume the existence of a predefined fuzzy partition set $\mathcal{MFS} = \{\mathcal{S}_i\}_{i=1}^F$, where $\mathcal{S}_i = \{\mathcal{A}_{ij}\}_{j=1}^{N_i}$ defines N_i fuzzy sets for the i -th feature. Each fuzzy set $\mathcal{A}_{ij}$ is associated with a membership function $\phi_{ij} : X_i \subseteq \mathbb{R} \rightarrow \mathbb{R}$, which maps the feature value x_i to its degree of membership in the fuzzy set.

Given a data point $\mathbf{x}_d = [x_{d1}, \dots, x_{dF}]$, we construct a graph instance $g_d = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes the node set and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges, as depicted in Fig. 2. The node set $\mathcal{V}$ is partitioned into two subsets:

- $\mathcal{V}_f = \{v_f = x_{di} \mid i = 1, \dots, F\}$, corresponding to raw feature nodes.
- $\mathcal{V}_s = \{v_s = \phi_{ij} \mid \phi_{ij} \in \mathcal{S}_i, \mathcal{S}_i \in \mathcal{MFS}\}$, corresponding to fuzzy set membership function nodes.

Each raw feature node x_{di} is connected to its associated fuzzy set nodes via edges, yielding the edge set:

$$\mathcal{E} = \{(v_f, v_s) \mid v_f = x_{di} \in \mathcal{V}_f, v_s = \phi_{ij} \in \mathcal{V}_s\}. \quad (1)$$

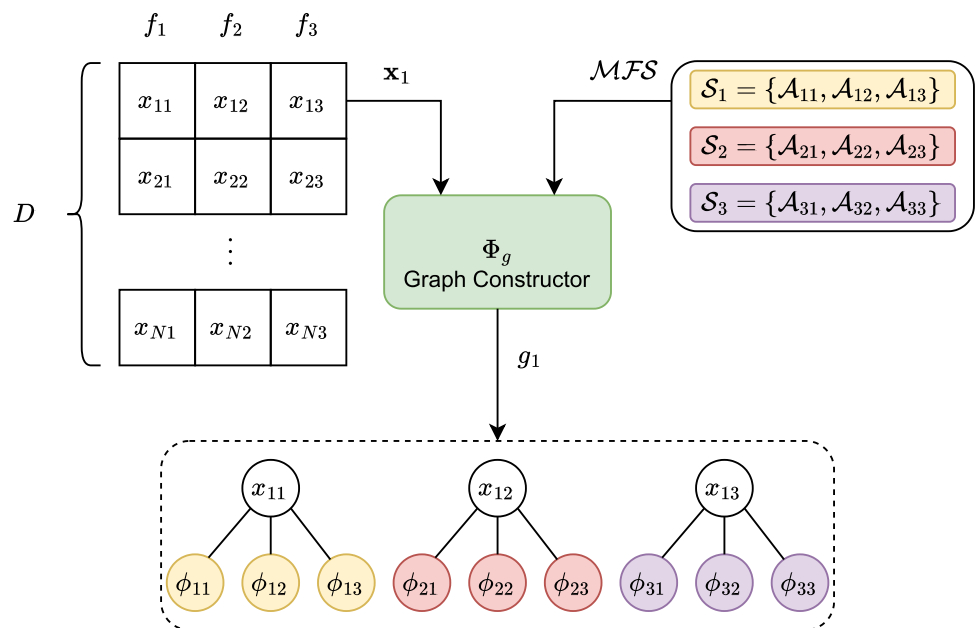
The graph g_d is further defined by a node feature matrix X and an adjacency matrix A . The node feature matrix $X = [x_1, \dots, x_{|\mathcal{V}|}] \in \mathbb{R}^{|\mathcal{V}| \times 1}$ provides a scalar feature for each node. For raw feature nodes in $\mathcal{V}_f$, the original feature value is retained. For fuzzy set nodes in $\mathcal{V}_s$, we propagate the corresponding raw input value x_{di} , allowing the activation of the membership function ϕ_{ij} during the forward pass. This enables differentiable optimization of fuzzy parameters via backpropagation (see Sect. 3.2). The adjacency matrix $A \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ encodes the graph topology, where $A_{ij} = 1$ if $(v_i, v_j) \in \mathcal{E}$ and $A_{ij} = 0$ otherwise.

The graph construction process can be formalized as a transformation function:

$$\Phi_g : \mathbb{R}^F \rightarrow \mathcal{G}, \quad \Phi_g(\mathbf{x}_d) = g_d, \quad (2)$$

which maps a tabular data point $\mathbf{x}_d$ to a graph instance $g_d \in \mathcal{G}$. Consequently, the original dataset D is converted into a new graph-based dataset $D_g = \{(g_d, y_d)\}_{d=1}^N$. Given F features and a total of $S = \sum_{i=1}^F |\mathcal{S}_i|$ fuzzy sets across all features, each graph g_d consists of $F + S$ nodes and $F \times S$ edges. This construction scales linearly with the number of features and introduces an explicit semantic structure by linking fuzzy sets to their associated features—enabling a symbiotic representation that blends numerical input and symbolic knowledge.

Fig. 2 Construction of a graph instance. Each data point is transformed into a graph in which nodes represent both the features and the membership functions of the corresponding fuzzy sets



3.2 Model architecture

FuzzyGNN processes each input graph g_d by converting the original classification task into a graph classification task that leverages both fuzzy logic and graph-based learning. As illustrated in Fig. 3, the architecture consists of two main components: the *Activation Module*, responsible for computing fuzzy membership values, and the *GNN Model*, which performs the final classification.

The Activation Module evaluates each fuzzy membership function ϕ_{ij} on the corresponding raw input x_{di} , producing a scalar activation p_{ij} . Specifically, for each node $v \in \mathcal{V}$, its feature $\mathbf{x}_v$ is computed as:

$$\mathbf{x}_v = \begin{cases} p_{ij} = \phi_{ij}(x_{di}) & \text{if } v \in \mathcal{V}_s, \\ x_{di} & \text{otherwise.} \end{cases} \quad (3)$$

The resulting activated graph is then fed into a generic GNN backbone. Each l -th layer of the GNN updates node embeddings using the standard message-passing formulation:

$$\mathbf{x}_i^{(l)} = \gamma^{(l)} \left(\mathbf{x}_i^{(l-1)}, \bigoplus_{j \in \mathcal{N}(i)} \omega^{(l)}(\mathbf{x}_i^{(l-1)}, \mathbf{x}_j^{(l-1)}) \right), \quad (4)$$

where $\mathbf{x}_i^{(l)}$ denotes the embedding of node i at layer l , $\mathcal{N}(i)$ is the neighborhood of node i , $\omega^{(l)}$ is a learnable message function, $\gamma^{(l)}$ is the update function, and $\bigoplus$ denotes a permutation-invariant aggregation (e.g., sum, mean, or max).

After L layers, the node embeddings $\mathbf{X}^{(L)} \in \mathbb{R}^{|\mathcal{V}| \times h}$ are aggregated into a graph-level representation by averaging:

$$\mathbf{x}_g = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \mathbf{x}_v^{(L)}. \quad (5)$$

This vector $\mathbf{x}_g \in \mathbb{R}^h$ is then passed through a linear classifier, producing the logit vector $\hat{\mathbf{y}} = [\hat{y}_1, \dots, \hat{y}_K]$ corresponding to class probabilities:

$$\Phi : \mathcal{G} \rightarrow \mathbb{R}^K, \quad \Phi(g_d) = \hat{\mathbf{y}}. \quad (6)$$

Notably, our model imposes no restriction on the shape of the membership functions, enabling support for various types, such as triangular, trapezoidal, or Gaussian sets. These functions are parameterized and optimized via gradient descent during training, making them fully differentiable.

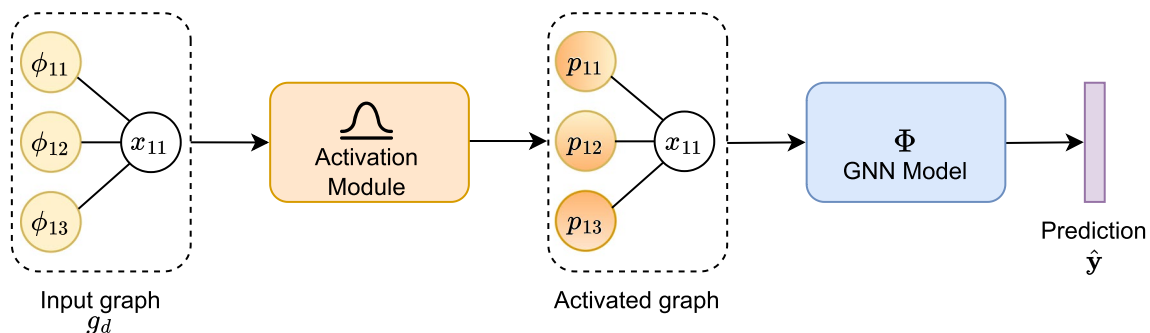


Fig. 3 FuzzyGNN main architecture. An input graph g_d is first processed by the Activation Module, which computes membership values based on the selected membership functions. The activated graph is then passed through the GNN Model, producing the predicted class probability vector $\hat{\mathbf{y}}$

Unlike traditional neuro-fuzzy systems, which often suffer from exponential growth in parameters with increasing feature dimensions, our architecture decouples parameter count from input size. The GNN maintains a fixed number of learnable parameters independent of the number of input features F , with only the fuzzy membership functions introducing F -dependent parameters. This design ensures scalability without compromising interpretability.

3.3 Rule extraction

To enhance interpretability, FuzzyGNN generates human-readable fuzzy rules for each prediction. Rule extraction is central to fuzzy models, as it consistently increases model transparency and trust. Furthermore, state-of-the-art approaches, such as ANFIS and other neuro-fuzzy models, typically include all descriptive features in the resulting rule. While exhaustive, this practice severely limits readability and interpretability, as users may be interested in only a subset of the most relevant features. To address these issues, our approach builds a dedicated *Explainer Module* that identifies the most influential features of the input graph and translates them into IF-THEN fuzzy rules, as illustrated in Fig. 4.

First, we leverage GNNExplainer [7] to identify the most important features for a given prediction. Specifically, given a graph $g_d = (\mathcal{V}, \mathcal{E})$ with node features X and adjacency matrix A , and a trained model Φ (Eq. (6)), GNNExplainer produces a node importance mask $\hat{X} \in [0, 1]^{1 \times F}$ and an edge importance mask $\hat{A} \in [0, 1]^{|\mathcal{V}| \times |\mathcal{V}|}$. These masks quantify the contribution of each feature and relation to the final output as:

$$\Phi_e : \mathcal{G} \rightarrow (\mathbb{R}^{1 \times F}, \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}), \quad \Phi_e(g_d) = (\hat{X}, \hat{A}). \quad (7)$$

Next, we select the most important features by thresholding $\hat{A}$ using a predefined value $\eta \in [0, 1]$. Here, η corresponds to the minimum importance percentile that an edge must reach in order to be included in the rule. Thus, the edge index matrix $\hat{A}$ is binarized as follows:

$$\hat{A}_{nu} = \begin{cases} 1 & \text{if } \hat{A}_{nu} \geq \eta, \\ 0 & \text{otherwise.} \end{cases} \quad (8)$$

Finally, we build the IF-THEN fuzzy rule by transforming the selected edges into symbolic terms. Recalling Eq. (1), each edge $e \in \mathcal{E}$ connects a feature to a fuzzy set. Thus, if the masked edge $\hat{A}_{nu} = 1$, the feature x_i referring to the node v_n is associated with the fuzzy set ϕ_{ij} , referring to the node v_u , from which we extract its linguistic term (e.g., *low*, *medium*, *high*). Selected edges belonging to different descriptive features form different antecedents, which are chained using the AND logic operator (e.g., “IF x_1 is *low* AND x_2 is *medium*...”). Important edges referring to the same feature are chained using the OR connector within a single antecedent term (e.g.,

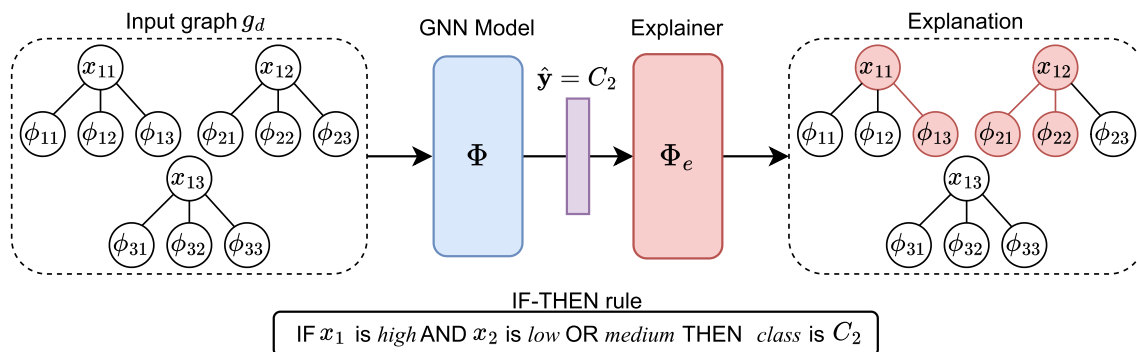


Fig. 4 Rule extraction process. The GNN Model processes the input graph g_d to obtain the predicted class probability distribution $\hat{y}$. The Explainer Module, utilizing GNNExplainer, identifies the most representative subgraph, which is then converted into a fuzzy rule

“IF x_5 is *medium* OR *high* AND x_{20} is *high*...”). The rule is concluded with the consequent, which corresponds to the predicted class.

The proposed mechanism allows flexible control depending on the choice of η . For example, smaller η values preserve more features—yielding broader rules—whereas larger values restrict the rule to only the strongest contributors. Significantly, we do not constrain the rules to include all features, but only those that exceed the relevance threshold, leading to concise and user-tailored explanations. An additional insight emerges by analyzing how the number of antecedents changes as η varies. This variation provides a direct view into the distribution of feature importance: if many antecedents disappear quickly as η increases, it indicates that only a small subset of features carries strong explanatory power, while the others contribute marginally. Conversely, a more gradual reduction in antecedents suggests a more balanced distribution of importance across features. This perspective not only enhances interpretability but also offers a novel way to characterize the robustness and sparsity of the model-derived explanations.

3.4 Fuzzy set pruning

Determining the optimal number of fuzzy sets per feature is a critical aspect of designing effective fuzzy systems. While one could rely on domain expertise to define fuzzy partitions manually, this approach introduces limitations, such as expert bias, limited generalizability, and unnecessary model complexity. In particular, using an excessive number of fuzzy sets can lead to overlapping or redundant membership functions, which degrade both predictive performance and interpretability.

To address these issues, FuzzyGNN incorporates a fuzzy set pruning mechanism, leveraging the factual explanations generated by the Explainer Module (see Sect. 3.3). After model training, we analyze validation data to identify and remove the least informative fuzzy sets. This process is driven by the edge importance masks returned by GNNExplainer (Eq. (7)), which quantify the relevance of each fuzzy set in the classification decision. By iteratively pruning low-impact fuzzy sets and retraining the model, we reduce complexity while preserving, or even improving, performance and transparency.

The pruning pipeline proceeds as follows:

1. Define a target validation metric ρ (e.g., cross-entropy loss, F1-score), a pruning percentile threshold τ , and a tolerance value δ to monitor performance degradation.
2. For each graph instance in the validation set, compute the edge importance mask using GNNExplainer, as described in Eq. (7).
3. For each membership function ϕ_{ij} , extract its relevance score from the edge mask $\hat{A}_{nu}$, where $e = (v_n, v_u)$ with $v_n = x_i$ (raw feature node) and $v_u = \phi_{ij}$ (fuzzy set node).
4. Average the relevance scores of each fuzzy set across all validation graphs to obtain a global importance score for every ϕ_{ij} .
5. Sort fuzzy sets by importance and retain only those above the τ percentile. If a feature has only one fuzzy set, remove the feature to avoid trivial rules.
6. Retrain the model Φ from scratch using the reduced fuzzy set configuration and evaluate the updated metric ρ on the validation set to obtain $r' = \rho(D_g^{val})$.
7. If $r' \geq r - \delta$, accept the new model and repeat from step (2). Otherwise, terminate and revert to the model from the previous iteration, which is considered optimal.

This pruning strategy yields a streamlined model that focuses on the most relevant fuzzy sets, improving both interpretability and efficiency. By reducing redundant or uninformative elements, FuzzyGNN can generate more concise and readable rules, overcoming a known limitation of state-of-the-art neuro-fuzzy approaches, where rule explosion often hampers model transparency and usability.

4 Experiments

In this section, we report the results of our experimental evaluation, which aims to assess both the predictive performance and interpretability of FuzzyGNN. We begin by describing the datasets employed in the study, followed by an overview of the baseline models used for comparison. We then outline the experimental protocol and provide implementation details to ensure reproducibility. Finally, we present both quantitative and qualitative results, as well as an in-depth efficiency analysis, highlighting the strengths and limitations of FuzzyGNN with respect to state-of-the-art approaches.

4.1 Datasets

Table 1 summarizes the datasets used in our evaluation. These datasets span a variety of application domains, including Biology, Health, Physics, Climate, and Computer Science, providing a diverse testbed for benchmarking both performance and interpretability. The datasets also vary in size: some contain hundreds of features (e.g., Gas Concentration), while others have fewer features (e.g., Ozone, Spambase). Five of the datasets are binary classification tasks, while Iris and Gas Concentration involve multiclass classification. All datasets were obtained from the UCI Machine Learning Repository.¹

Before training, we conducted a uniform preprocessing pipeline to ensure comparability across datasets. Non-informative features—such as categorical or textual variables—were removed, as fuzzy logic requires numerical inputs to define granular membership functions. For datasets containing missing values, we performed mean imputation on a per-feature basis. Finally, each dataset was partitioned into three subsets: 70% for training, 20% for validation, and 10% for testing, ensuring a stratified split where applicable.

4.2 Baselines

We compared our proposed method with nine baseline machine learning models selected to represent a spectrum of capabilities in terms of accuracy and transparency. These baseline models are grouped into three categories: *neuro-fuzzy*, *interpretable*, and *opaque* (black-box). Most models were implemented using the `scikit-learn` library² to ensure consistency in preprocessing and evaluation. Neuro-fuzzy baselines were developed following the implementation provided in [26].

We deliberately included well-established, interpretable, and neuro-fuzzy models in our comparison, as they remain the most widely adopted solutions in domains that require explainability. In contrast, many modern deep

Table 1 Statistics of the datasets used in the experimental phase

Dataset	Domain	#Inst.	#Feat.	#Used	#Class
Iris [29]	Biology	150	4	4	3
Cancer [30]	Health	569	30	30	2
Haberman [31]	Health	306	3	3	2
Heart disease [32]	Health	1190	11	5	2
Gas concentration [33]	Climate	13,910	129	129	6
Ozone [34]	Climate	2536	72	72	2
Spambase [35]	Comp. sci	4601	57	57	2

For each dataset, we report its domain of application, the total number of instances, the number of original features, the number of features retained after preprocessing, and the number of target classes

¹ UCI repository: <https://archive.ics.uci.edu/>

² Scikit-learn: <https://scikit-learn.org/stable/>

learning methods, while powerful, lack the interpretability constraints that FuzzyGNN is explicitly designed to address.

Neuro-fuzzy models We included three neuro-fuzzy systems. The first is ANFIS [36], which combines neural networks with fuzzy logic to learn fuzzy rules from data while maintaining interpretability through IF-THEN structures. The second is the Takagi–Sugeno–Kang-based model (TSK [37]), which addresses the curse of dimensionality commonly encountered in fuzzy systems. Unlike ANFIS, however, TSK does not rely on linguistic variables but instead identifies distinct fuzzy set combinations for each rule, thereby reducing human interpretability. Finally, we established a strong FuzzyGNN baseline. Inspired by PLENARY [28], we first trained a Multilayer Perceptron (MLP), then applied SHAP [10] to rank the input features. Subsequently, following Wang et al. [38], we selected the top 80% most influential features and trained FuzzyGNN exclusively on this feature-refined version of each dataset.

Interpretable models We also considered three models known for their transparency. Decision Trees (DT) [39] are rule-based classifiers whose hierarchical structure is inherently interpretable. Logistic Regression (LR) [40] offers interpretability through linear coefficients, although this clarity may diminish with high-dimensional or interaction-rich data. Naive Bayes (NB) [41], a simple probabilistic model that assumes feature independence, is often favored for its simplicity and interpretability, particularly in domains involving text or categorical data.

Opaque models For comparison with high-performing but non-transparent approaches, we included Random Forests (RF) [42] and XGBoost (XGB) [43]. While based on decision trees, their ensemble structure hinders interpretability. We also evaluated Support Vector Machines (SVM) [44], which offer strong generalization but lack transparency—especially when using nonlinear kernels. Finally, the Multilayer Perceptron [45], a deep feedforward neural network, was included as a typical black-box model with high capacity but minimal interpretability.

4.3 Experimental setup

In this study, we evaluated the effectiveness of FuzzyGNN by modeling fuzzy sets using Gaussian membership functions. Each fuzzy set $\mathcal{A}_{ij}$ was associated with a membership function defined as:

$$p_{ij} = \phi_{ij}(x_{di}) = \exp\left(-\frac{(x_{di} - c_{ij})^2}{\sigma_{ij}^2}\right), \quad (9)$$

where c_{ij} and σ_{ij} denote the center and width of the Gaussian function, respectively.

To initialize the membership functions, we assumed each feature i has a domain $X_i = [a_i, b_i] \subseteq \mathbb{R}$. Given N_i fuzzy sets for feature i , the initial parameters were computed as:

$$c_{ij} = a_i + (j - 1) \frac{b_i - a_i}{N_i - 1}, \quad (10)$$

$$\sigma_{ij} = \frac{b_i - a_i}{N_i}, \quad (11)$$

for $j = 1, \dots, N_i$. These values ensure that the fuzzy sets uniformly cover the input space. While Gaussian functions were used in our main experiments, the framework is compatible with any parametric membership function (e.g., triangular or trapezoidal), providing flexibility for modeling uncertainty.

It is important to note that selecting the appropriate number of fuzzy sets for each feature is a critical decision in fuzzy modeling. Following the approach of Guillaume et al. [46], we decided to model five fuzzy sets for each input feature. A smaller number may fail to adequately capture nonlinear behaviors, while an excessively large number of fuzzy sets could lead to overly specific representations, reducing the model's generality.

The GNN backbone was tuned using a Tree-Structured Parzen Estimator (TPE) for Bayesian optimization. The following hyperparameters were explored:

- Number of GNN layers $L \in \{1, 2, 3\}$,
- Aggregation function $\oplus \in \{\text{sum}, \text{min}, \text{max}\}$,
- Message-passing layer type: GATv2 [47], ResGatedGraphConv [48], or GCNConv [49],
- Initial learning rate $\in \{10^{-3}, 10^{-4}, 10^{-5}\}$.

The best configuration was selected by minimizing the validation loss on the development set D_g^{val} .

The model was trained using the AdamW optimizer [50] to minimize the categorical cross-entropy loss. Each run was trained for a maximum of 150 epochs, with a learning rate reduced by a factor of 0.1 after 5 epochs without validation improvement (with a cooldown period of 2 epochs). Early stopping was triggered if no improvement was observed for 20 consecutive epochs to avoid overfitting.

Training followed the pipeline described in Sect. 3.2, with a fixed batch size of 128. All experiments were conducted on a machine equipped with 64 GB of RAM and an NVIDIA GeForce RTX 3080 GPU (12 GB VRAM).

4.4 Evaluation metrics

We conducted extensive experiments to evaluate both the predictive performance and interpretability of the proposed model. For the quantitative assessment, we report standard classification metrics—accuracy, F1-score, precision, and recall—to provide a comprehensive performance overview. To assess interpretability, we adopt a multi-level evaluation strategy. At the model level, following the guidelines of [17], we consider indicators such as the number of input features. At the fuzzy set level, we include metrics such as the Jaccard structural similarity, external distance between fuzzy sets, the number of linguistic terms, and the total number of fuzzy parameters. At the rule level, we report the number of extracted rules and the average number of antecedents per rule, providing insight into the rules' compactness and clarity.

Specifically, the Jaccard structural similarity between two fuzzy sets measures their overlap and is defined as:

$$Jacc_S(\mathcal{A}, \mathcal{B}) = \frac{\text{area}(\mathcal{A} \otimes \mathcal{B})}{\text{area}(\mathcal{A} \oplus \mathcal{A})}, \quad (12)$$

where $\mathcal{A}$ and $\mathcal{B}$ denote two fuzzy sets, $\otimes$ and $\oplus$ are the fuzzy intersection and union operators, and $\text{area}(\mathcal{A}) = \int_{-\infty}^{+\infty} \mathcal{A}(x) dx$. Averaging the Jaccard index across all features and all fuzzy set pairs yields:

$$Jacc(D) = \frac{1}{F} \sum_{i=1}^F Jacc_F(\mathcal{S}_i), \quad Jacc_F(\mathcal{S}_i) = \frac{1}{|\mathcal{S}_i^2|} \sum_{(\mathcal{A}, \mathcal{B}) \in \mathcal{S}_i^2} Jacc_S(\mathcal{A}, \mathcal{B}), \quad (13)$$

where values close to 1 indicate high redundancy and values near 0 suggest distinct sets; intermediate values are ideal for interpretability.

The external distance between fuzzy sets $\mathcal{A}$ and $\mathcal{B}$, defined on the domain of feature f_i , is:

$$Dm_{ext}(\mathcal{A}, \mathcal{B}, \mathbf{p}, \mathbf{q}) = \frac{1}{N} \sum_{j=1}^N (|\mathcal{A}(p_j) - \mathcal{B}(q_j)| + Dm_{prot}(\mathcal{A}, \mathcal{B})), \quad (14)$$

where $Dm_{prot}(\mathcal{A}, \mathcal{B})$ denotes the distance between set centers and $\mathbf{p}, \mathbf{q} \in \mathbb{R}^N$ are vectors sampled from D_{test} . Averaging over all feature pairs, we define:

$$Dm(D) = \frac{1}{F} \sum_{i=1}^F Dm_f(\mathcal{S}_i, \mathbf{p}_i, \mathbf{q}_i), \quad Dm_f(\mathcal{S}_i, \mathbf{P}, \mathbf{Q}) = \frac{1}{|\mathcal{S}_i^2|} \sum_{(\mathcal{A}, \mathcal{B}) \in \mathcal{S}_i^2} Dm_{ext}(\mathcal{A}, \mathcal{B}, \mathbf{P}_i, \mathbf{Q}_i). \quad (15)$$

Table 2 Classification accuracy across datasets

Model	Iris	Cancer	Haberman	Heart disease	Gas concentration	Ozone	Spambase
ANFIS	0.91	0.92 [†]	0.70	0.77	0.68 [†]	0.94 [†]	0.85 [†]
TSK	0.86	0.95	0.51	0.74	0.98	0.94	0.93
SHAP-FuzzyGNN	0.88	0.94	0.73	0.70	0.88	0.93	0.84
DT	0.87	0.93	0.67	0.76	0.97	0.93	0.90
LR	0.80	0.96	0.74	0.75	0.97	0.95	0.91
NB	0.91	0.93	0.69	0.73	0.58	0.68	0.82
RF	0.91	0.92	0.70	0.75	0.99	0.94	0.95
XGB	0.93	0.91	0.74	0.77	0.99	0.94	0.93
SVM	0.91	0.97	0.70	0.75	0.85	0.94	0.92
MLP	0.91	0.98	0.67	0.76	0.99	0.93	0.94
FuzzyGNN	0.93	0.95	0.74	0.75	0.95	0.94	0.88

[†] indicates a revised ANFIS version with feature selection. The full metric breakdown is reported in Appendix A

4.5 Quantitative results

We first assess the predictive performance of FuzzyGNN relative to the baselines outlined in Sect. 4.2, focusing on accuracy. Table 2 reports the results across all datasets. A complete set of evaluation metrics (precision, recall, F1-score) is provided in Appendix A.

Due to computational constraints, ANFIS could not be trained on high-dimensional datasets without prior feature selection. To address this, we applied an ANOVA-based feature selection step, retaining 3–9 variables while keeping five fuzzy sets per feature. We then selected the best-performing configuration for each dataset. This enhanced ANFIS version is marked with [†] in the table.

Overall, FuzzyGNN achieves competitive accuracy across most datasets. On Iris, it reaches 93% accuracy—matching XGB and outperforming most interpretable models. On Cancer, it obtains 95%, which is only marginally lower than SVM (97%) and MLP (98%), yet higher than most interpretable alternatives. On Ozone, it matches the best-performing models (94%), while on Spambase it achieves 88%, trailing RF and MLP but still surpassing ANFIS. On Haberman, FuzzyGNN achieves 74%, aligning with LR and XGB, and improving over ANFIS by 4 points. On Heart Disease, its performance (75%) is very close to most baselines, though slightly below ANFIS (77%). Finally, on the more challenging Gas Concentration dataset, FuzzyGNN obtains 95%, which, while lower than ensemble methods and MLP (all near 99%), is a substantial improvement over ANFIS (68%).

The comparison with ANFIS is particularly relevant, as both models are designed to produce interpretable fuzzy IF-THEN rules. Unlike ANFIS, which requires feature selection to remain computationally feasible and tends to include all features in every rule, FuzzyGNN operates directly on the whole feature space while maintaining a constant parameter count relative to input dimensionality. This allows for more effective scaling to high-dimensional datasets. Notably, FuzzyGNN consistently surpasses ANFIS on various benchmarks (e.g., Iris, Haberman, Gas Concentration) while achieving similar accuracy on others.

Furthermore, FuzzyGNN consistently achieves better performance than its baseline SHAP-FuzzyGNN across all datasets. This outcome quantitatively underscores the importance of our framework's internal feature selection stage in effectively identifying and utilizing only the essential features.

These results demonstrate that FuzzyGNN can balance interpretability and accuracy across diverse tabular classification tasks. Although it does not always match the performance of the most powerful ensembles or deep models, it offers a clear advantage over traditional neuro-fuzzy systems by combining scalability with concise, human-readable rule extraction.

Table 3 Training and inference times (mean $\pm$ std), computed on batches of 128 data points and reported in milliseconds

Model	Iris		Cancer		Haberman		Heart disease		Gas concentration		Ozone		Spambase		Avg	
# Features	# 4		# 30		# 3		# 5		# 129		# 72		# 57			
ANFIS	8.99 $\pm$ 0.00		147.15 $\pm$ 26.65 [†]		5.11 $\pm$ 0.33		13.96 $\pm$ 0.71		8.90 $\pm$ 0.12 [†]		27.27 $\pm$ 66.96 [†]		156.11 $\pm$ 12.67 [†]		52.5	
	2.70 $\pm$ 0.00		61.36 $\pm$ 0.00 [†]		1.83 $\pm$ 0.00		7.02 $\pm$ 0.00		3.07 $\pm$ 0.08 [†]		2.52 $\pm$ 0.00 [†]		128.59 $\pm$ 25.61 [†]		29.58	
TSK	1.67 $\pm$ 0.00		1.45 $\pm$ 0.10		1.68 $\pm$ 0.00		1.40 $\pm$ 0.05		1.38 $\pm$ 0.04		1.38 $\pm$ 0.04		1.52 $\pm$ 0.05		1.5	
	2.57 $\pm$ 0.00		0.35 $\pm$ 0.00		0.35 $\pm$ 0.00		0.32 $\pm$ 0.00		0.33 $\pm$ 0.02		0.30 $\pm$ 0.04		0.29 $\pm$ 0.04		0.64	
DT	0.49 $\pm$ 0.00		0.55 $\pm$ 0.06		0.44 $\pm$ 0.03		0.55 $\pm$ 0.03		0.77 $\pm$ 0.07		0.49 $\pm$ 0.05		0.48 $\pm$ 0.03		0.54	
	0.12 $\pm$ 0.00		0.13 $\pm$ 0.00		0.12 $\pm$ 0.00		0.14 $\pm$ 0.00		0.16 $\pm$ 0.03		0.15 $\pm$ 0.00		0.13 $\pm$ 0.01		0.14	
LR	1.73 $\pm$ 0.00		0.63 $\pm$ 0.05		1.23 $\pm$ 0.06		0.66 $\pm$ 0.06		16.49 $\pm$ 2.46		12.72 $\pm$ 1.68		2.39 $\pm$ 0.33		5.12	
	0.12 $\pm$ 0.00		0.12 $\pm$ 0.00		0.11 $\pm$ 0.00		0.13 $\pm$ 0.00		0.13 $\pm$ 0.01		1.62 $\pm$ 0.00		0.14 $\pm$ 0.02		0.34	
NB	0.63 $\pm$ 0.00		0.57 $\pm$ 0.01		0.57 $\pm$ 0.03		0.55 $\pm$ 0.02		0.82 $\pm$ 0.03		0.61 $\pm$ 0.04		0.58 $\pm$ 0.02		0.62	
	0.16 $\pm$ 0.00		0.17 $\pm$ 0.00		0.15 $\pm$ 0.00		0.17 $\pm$ 0.00		0.47 $\pm$ 0.01		0.21 $\pm$ 0.00		0.18 $\pm$ 0.02		0.22	
RF	520.09 $\pm$ 0.00		36.88 $\pm$ 0.36		36.31 $\pm$ 0.35		227.14 $\pm$ 12.68		1163.43 $\pm$ 34.78		545.14 $\pm$ 6.24		982.51 $\pm$ 24.14		501.64	
	20.21 $\pm$ 0.00		1.66 $\pm$ 0.00		1.61 $\pm$ 0.00		8.87 $\pm$ 0.00		39.71 $\pm$ 0.75		21.02 $\pm$ 0.00		37.98 $\pm$ 0.80		18.72	
XGB	59.25 $\pm$ 0.00		33.30 $\pm$ 0.92		29.03 $\pm$ 0.39		23.73 $\pm$ 0.16		341.82 $\pm$ 27.52		31.20 $\pm$ 0.15		59.46 $\pm$ 0.82		82.54	
	0.75 $\pm$ 0.00		0.77 $\pm$ 0.00		0.69 $\pm$ 0.00		0.75 $\pm$ 0.00		1.25 $\pm$ 0.11		0.68 $\pm$ 0.00		0.78 $\pm$ 0.15		0.81	
SVM	0.96 $\pm$ 0.00		1.20 $\pm$ 0.22		1.38 $\pm$ 0.73		1.53 $\pm$ 0.29		4.58 $\pm$ 0.48		1.96 $\pm$ 0.41		2.98 $\pm$ 0.37		2.08	
	0.15 $\pm$ 0.00		0.21 $\pm$ 0.00		0.16 $\pm$ 0.00		0.25 $\pm$ 0.00		0.24 $\pm$ 0.02		0.29 $\pm$ 0.00		0.42 $\pm$ 0.07		0.25	
MLP	176.21 $\pm$ 0.00		336.40 $\pm$ 234.31		200.21 $\pm$ 79.23		82.43 $\pm$ 5.54		450.59 $\pm$ 109.90		161.12 $\pm$ 109.17		227.02 $\pm$ 60.83		233.43	
	0.25 $\pm$ 0.00		0.44 $\pm$ 0.00		0.29 $\pm$ 0.00		0.28 $\pm$ 0.00		1.19 $\pm$ 0.06		0.40 $\pm$ 0.00		0.34 $\pm$ 0.04		0.46	
FuzzyGNN	4.82 $\pm$ 0.00		47.84 $\pm$ 2.32		5.63 $\pm$ 0.99		13.80 $\pm$ 0.38		245.68 $\pm$ 45.36		144.95 $\pm$ 7.33		60.37 $\pm$ 10.95		74.73	
	4.27 $\pm$ 0.00		17.25 $\pm$ 0.00		3.30 $\pm$ 0.00		4.65 $\pm$ 0.00		60.78 $\pm$ 1.36		47.45 $\pm$ 1.50		18.67 $\pm$ 1.09		22.34	

[†] indicates a revised ANFIS version with feature selection

4.6 Efficiency analysis

In this section, we analyze the computational complexity of our model. Table 3 reports the training and validation times of all the evaluated methodologies across datasets. A first observation is that most classical machine learning algorithms require comparatively lower computation time. However, it is important to emphasize that, although all timings were measured on the same hardware (as described in Sect. 4.3), some models exploit GPU acceleration (e.g., ANFIS, TSK, FuzzyGNN), whereas others rely exclusively on CPUs (e.g., DT, LR, NB). Despite this heterogeneity, FuzzyGNN consistently achieves better efficiency than RF, XGB, and MLP across datasets. Conversely, our model is slightly less efficient than ANFIS. Nonetheless, for high-dimensional datasets, we were required to reduce the input feature space in advance to address ANFIS's computational constraints (i.e., ANFIS[†]).

Overall, our results indicate that the computational cost of our model grows superlinearly with the number of features. This trend aligns with the time complexity typically associated with GNNs [51], which can be expressed as:

$$O_t(\Phi) \approx O\left(L \cdot |\mathcal{E}| \cdot h + L \cdot |\mathcal{V}| \cdot h^2\right), \quad (16)$$

where L denotes the total number of graph layers, $\mathcal{V}$ and $\mathcal{E}$ represent the sets of nodes and edges, respectively, and h is the dimensionality of the hidden node features.

Nevertheless, we show that the training and inference overhead introduced by our framework is mitigated by the nearly constant number of learnable parameters, which remains largely independent of the input feature dimensionality. Table 4 reports the number of parameters learned by each model across datasets. In this regard, the number of parameters of FuzzyGNN depends on feature dimensionality only within the fuzzification layer, whose learnable parameters scale as:

$$O_{am} = O(F \times k_m), \quad (17)$$

where O_{am} is the space complexity of the activation module of FuzzyGNN, F is the number of input dataset features, and k_m denotes the number of learnable parameters associated with the selected membership function (e.g., μ and σ^2 for a Gaussian). By contrast, the number of parameters in the GNN backbone [51] is completely independent of the dataset's input dimensionality, as it follows:

$$O_\Phi = O\left(L \cdot (h^2 + o(GNN))\right), \quad (18)$$

where $o(GNN)$ accounts for the additional parameters introduced by specific GNN layer types (e.g., GCN, Res-Gated, GAT). Moreover, the parameter count of the activation module is substantially smaller than that of the backbone, and therefore negligible in practice. As a result, unlike other neural baselines such as ANFIS or MLP, our model maintains a nearly constant parameter growth with respect to feature dimensionality, effectively bridging the gap toward lightweight neuro-fuzzy systems.

4.7 Qualitative results

While FuzzyGNN does not consistently achieve the highest classification accuracy, the pruning phase simplifies the model and enhances interpretability without compromising performance. Accordingly, we conducted a comparative evaluation, benchmarking FuzzyGNN against ANFIS, TSK, and its strong baseline SHAP-FuzzyGNN, using the metrics described in Sect. 4.4.

Tables 5 and 6 summarize the comparison. FuzzyGNN consistently uses fewer features than ANFIS and TSK, thanks to its fuzzy pruning mechanism. In contrast, SHAP-FuzzyGNN eventually employs fewer input

Table 4 Number of parameters learned for each model across datasets

Model	Iris	Cancer	Haberman	Heart disease	Gas concentration	Ozone	Spambase
# Features	# 4	# 30	# 3	# 5	# 129	# 72	# 57
ANFIS	1915	156320 [†]	72	6300	3790 [†]	280 [†]	156320 [†]
TSK	133	642	1302	3082	69166	7402	9442
DT	15	43	3	219	711	29	575
LR	15	31	4	6	780	73	58
NB	15	62	8	12	780	146	116
RF	3750	150	1792	2266	453124	4252	334404
XGB	1440	494	480	398	4918	480	1016
SVM	189	1696	520	3759	238410	16946	55165
MLP	5503	6701	10401	701	16956	13851	13101
FuzzyGNN	67463	271010	270854	67366	68534	67870	67544

[†] refers to a revised ANFIS version with feature selection

Table 5 Interpretability metrics across datasets, benchmarking FuzzyGNN against its simplified version SHAP-FuzzyGNN, ANFIS, and TSK

Dataset	Model	# features	# fuzzy parameters	# extracted rules	Avg. # antecedents $\eta \in \{0.25, 0.50, 0.75\}$
Iris	ANFIS	4	40	625	4
	TSK	4	80	10	4
	SHAP-FuzzyGNN	3	30	1	(3–2.31–1.57)
	FuzzyGNN	1	4	1	(1–1–1)
Cancer	ANFIS [†]	7	70	78,125	7
	TSK	24	120	10	24
	SHAP-FuzzyGNN	24	240	1	(9.98–6.52–4.79)
	FuzzyGNN	24	160	1	(9.99–6.99–4.62)
Haberman	ANFIS	3	30	125	3
	TSK	3	780	130	3
	SHAP-FuzzyGNN	2	20	1	(2–2–2)
	FuzzyGNN	1	4	1	(1–1–1)
Heart disease	ANFIS	5	50	3125	5
	TSK	5	2200	220	5
	SHAP-FuzzyGNN	4	40	1	(3.98–2.01–1.58)
	FuzzyGNN	5	36	1	(4.94–2.95–2.48)
Gas concentration	ANFIS [†]	4	40	625	4
	TSK	129	67,080	260	129
	SHAP-FuzzyGNN	57	1,030	1	(10–9.90–8.10)
	FuzzyGNN	106	344	1	(10–9.97–7.71)
Ozone	ANFIS [†]	3	30	125	3
	TSK	72	7200	50	72
	SHAP-FuzzyGNN	57	570	1	(10–9.96–7.88)
	FuzzyGNN	72	540	1	(10–9.94–6.51)
Spambase	ANFIS [†]	7	70	78,125	7
	TSK	40	6400	80	40
	SHAP-FuzzyGNN	45	450	1	(10–7.34–6.37)
	FuzzyGNN	40	214	1	(9.97–6.67–3.87)

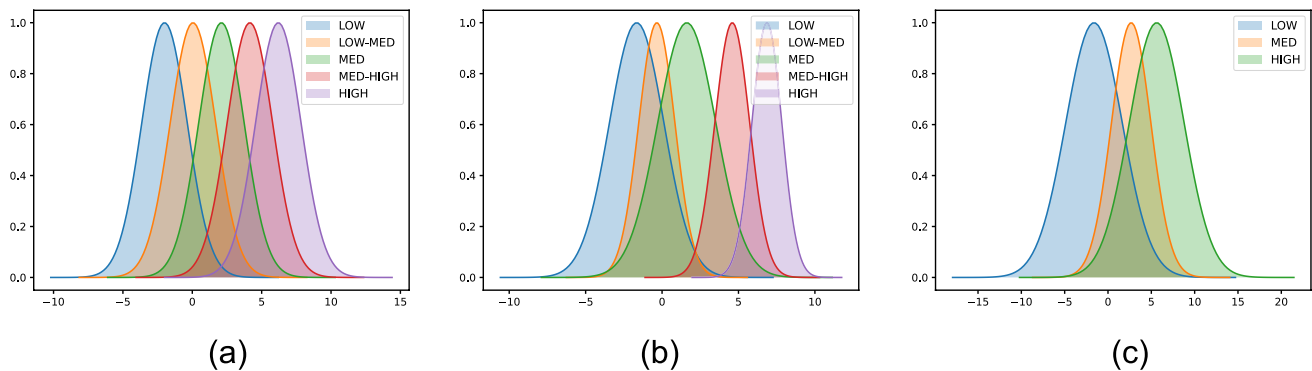
Metrics include the number of features, fuzzy parameters, rules, and antecedents. [†] indicates a revised ANFIS version

features than FuzzyGNN. However, this is because we systematically retained only 80% of the input features. In FuzzyGNN, instead, feature selection is performed dynamically, allowing the model to identify the most relevant inputs for each dataset. Additionally, the number of fuzzy parameters is drastically reduced, especially compared to TSK, which assigns different fuzzy sets to each rule and becomes impractical on large datasets. Whereas neuro-fuzzy baselines typically produce, after training, a full FIS composed of an exponential number of

Table 6 Fuzzy set interpretability metrics: Jaccard index, external distance, and linguistic terms, benchmarking FuzzyGNN against its simplified baseline SHAP-FuzzyGNN and ANFIS

Dataset	Model	Jaccard index	External distance	# linguistic terms
Iris	ANFIS	1e−4	0.61	20
	SHAP-FuzzyGNN	0.25	1.80	15
	FuzzyGNN	0.46	2.66	2
Cancer	ANFIS [†]	2e−3	0.79	35
	SHAP-FuzzyGNN	0.38	48.14	120
	FuzzyGNN	0.47	164.24	80
Haberman	ANFIS	4e−3	0.60	15
	SHAP-FuzzyGNN	0.18	13.91	10
	FuzzyGNN	0.51	43.26	2
Heart disease	ANFIS	2e−3	0.65	25
	SHAP-FuzzyGNN	0.22	123.45	20
	FuzzyGNN	0.35	114.70	18
Gas concentration	ANFIS [†]	4e−4	0.97	20
	SHAP-FuzzyGNN	0.14	7.0e4	515
	FuzzyGNN	0.21	1.6e4	344
Ozone	ANFIS [†]	2e−5	0.69	15
	SHAP-FuzzyGNN	0.25	23.87	285
	FuzzyGNN	0.32	31.03	270
Spambase	ANFIS [†]	0.02	0.99	35
	SHAP-FuzzyGNN	0.11	124.38	225
	FuzzyGNN	0.17	401.31	107

[†] indicates a revised ANFIS version

**Fig. 5** Membership functions learned by ANFIS **a**, and by FuzzyGNN before **b** and after **c** pruning

rules—each involving all descriptive features—FuzzyGNN extracts a single rule at inference time, tailored to the given test instance, thereby enhancing transparency and human interpretability. Thanks to the Explainer Module (see Sect. 3.3), we also limit the number of antecedents in the generated fuzzy rule, further improving readability.

To assess this aspect, we compared the mean number of antecedents included in the output fuzzy rule in the test set for different η values, namely $\eta \in \{0.25, 0.50, 0.75\}$. These correspond, respectively, to rules including the top Q3 features ($\eta = 0.25$), top Q2 features ($\eta = 0.50$), and top Q1 features ($\eta = 0.75$). Therefore, unlike prior state-of-the-art neuro-fuzzy systems, our rule extraction process can also provide an indicator of the skewness of the feature-importance distribution for a given dataset.

Additionally, the superiority of FuzzyGNN in fuzzy set design is also proved by fuzzy set metrics, as illustrated in Table 6. The Jaccard index is significantly higher than both ANFIS and SHAP-FuzzyGNN, indicating more meaningful overlap and better modeling of uncertainty. Similarly, the external distance is larger, reflecting greater separability and distinctiveness of fuzzy terms. This is visually illustrated in Fig. 5, which shows the fuzzy sets learned for the feature *old peak* in the Heart Disease dataset. ANFIS (a) generates poorly distributed terms, with

Table 7 Examples of human-interpretable fuzzy rules produced by FuzzyGNN for each dataset, illustrating the model's ability to capture meaningful patterns through concise linguistic expressions

Dataset	Rule
Iris	IF <i>petal_length</i> is <i>low</i> THEN <i>class</i> is <i>Iris_Virginica</i>
Cancer	IF <i>concavity_mean</i> is <i>high</i> and <i>simmetry_worst</i> is <i>low</i> and <i>concave_points_mean</i> is (<i>medium</i> or <i>high</i>) THEN <i>class</i> is <i>positive</i>
Haberman	IF <i>age</i> is <i>low</i> THEN <i>class</i> is <i>survivor</i>
Heart disease	IF <i>cholesterol</i> is (<i>med_high</i> or <i>high</i>) and <i>age</i> is <i>medium</i> and <i>resting_bp</i> is <i>medium</i> and <i>max_heart_rate</i> is (<i>med_high</i> or <i>high</i>) THEN <i>class</i> is <i>heart_disease</i>
Gas concentration	IF <i>F6</i> is <i>low</i> and <i>F10</i> is <i>med-high</i> and <i>F100</i> is (<i>med</i> or <i>high</i>) and <i>F0</i> is <i>high</i> THEN <i>class</i> is <i>ammonia</i>
Ozone	IF <i>temp_peak</i> is <i>high</i> and <i>temp_average</i> is (<i>medium</i> or <i>high</i>) and <i>U85</i> is <i>low</i> and <i>V85</i> is <i>low</i> and <i>temp_50</i> is <i>high</i> and <i>K_index</i> is <i>medium</i> THEN <i>class</i> is <i>ozone_day</i>
Spambase	IF <i>wf_you</i> is (<i>med</i> or <i>high</i>) and <i>wf_recieve</i> is <i>high</i> THEN <i>class</i> is <i>spam</i>

redundancy in the ‘high’ region. In contrast, FuzzyGNN (b) learns evenly spaced fuzzy sets, while pruning (c) reduces the number of terms to only two, enhancing simplicity and semantic clarity.

Thus, by reducing fuzzy terms while maintaining performance, FuzzyGNN provides interpretable models with fewer fuzzy parameters and cleaner rule structures. These characteristics make it ideal for sensitive application domains, such as healthcare or finance. Compared to ANFIS[†] and TSK-based models, FuzzyGNN achieves high interpretability with fewer features, parameters, and rules—without sacrificing accuracy—thus paving the way for practical, trustworthy, and explainable AI.

To illustrate the quality of the extracted rules, Table 7 presents a representative fuzzy rule generated by FuzzyGNN for each dataset. The rules are generally simple, coherent, and semantically meaningful—hallmarks of interpretability. In the Cancer dataset, the rule relies on three fuzzy variables: one described by the linguistic term *high*, one by *low*, and one by either *medium* OR *high*. Notably, the model retains only a limited subset of input features, yielding a compact, readable rule that accurately predicts the correct class. In the Haberman dataset, a single variable—*age*—is sufficient to formulate the rule. When *age* is *low*, the model predicts survival. Although a univariate rule may oversimplify a complex medical condition like cancer, this still reflects the discriminative role of age in the dataset. In the Heart Disease dataset, the rule combines multiple variables with diverse linguistic terms. Notably, the *cholesterol* variable activates either the *med_high* or *high* term via a logical OR, aligning with clinical knowledge that elevated cholesterol is a risk factor for cardiovascular issues. Additional variables, such as *age* (*medium*) and *max_heart_rate* (*med_high* or *high*), reinforce the coherence and plausibility of the prediction. Together, these rule components capture realistic correlations between physiological indicators and heart disease, underscoring the model's ability to generate meaningful, interpretable knowledge.

This interpretability can be extended to other domains as well. For instance, in blood analysis, features such as *hemoglobin level*, *white blood cell count*, and *platelet count* would be associated with fuzzy terms. Subsequently, by selecting the most relevant blood parameters and mapping them to their respective linguistic terms, FuzzyGNN could provide interpretable rules indicating that a combination of *low hemoglobin level* and *high white blood cell count* implies *anemia*. As a result, FuzzyGNN could provide a concrete use case in which end users (i.e., doctors) can be substantially helped in delivering diagnoses by identifying the most relevant information, in combination with granularized feature values.

5 Conclusion

This paper introduced FuzzyGNN, a hybrid framework that integrates fuzzy logic with Graph Neural Networks for interpretable classification on tabular data. Unlike conventional neuro-fuzzy models, which often suffer from scalability and redundancy issues, FuzzyGNN leverages graph-based representations to model contextual relationships among features and fuzzy sets while maintaining a compact and transparent rule base.

FuzzyGNN provides both interpretability and explainability: it is interpretable by design, thanks to the compact fuzzy rule base derived from the graph structure, and explainable through post-hoc analysis enabled by GNNExplainer. This dual capacity enhances user trust and facilitates model auditing, which is essential in critical domains such as healthcare and finance.

Experimental results show that FuzzyGNN achieves competitive accuracy across various datasets, often matching or outperforming traditional interpretable models. More importantly, it significantly improves interpretability through a pruning mechanism that reduces the number of fuzzy sets, rules, and antecedents without compromising performance. Compared to state-of-the-art neuro-fuzzy baselines such as ANFIS and TSK variants, FuzzyGNN produces simpler models with fewer parameters and more consistent fuzzy partitions—making it a viable solution in domains where interpretability is essential.

Future work will explore extending the model to handle larger and more heterogeneous datasets, as well as investigating its integration into decision support systems involving human-in-the-loop scenarios. Further developments may also include incorporating categorical features into our framework, enabling it to handle diverse and heterogeneous information.

Appendix A

This appendix provides supplementary results from the quantitative evaluation of the proposed model. Specifically:

- Table 8 reports the F1-score obtained across all datasets.
- Table 9 presents the corresponding Precision values.
- Table 10 shows the recall values.

Table 8 F1-score results across datasets

Model	Iris	Cancer	Haberman	Heart disease	Gas concentration	Ozone	Spambase
ANFIS	0.91	0.92 [†]	0.52	0.77	0.61 [†]	0.48 [†]	0.83 [†]
TSK	0.86	0.95	0.48	0.74	0.98	0.71	0.92
SHAP-FuzzyGNN	0.89	0.94	0.42	0.70	0.88	0.50	0.84
DT	0.87	0.92	0.54	0.76	0.97	0.52	0.90
LR	0.79	0.96	0.42	0.75	0.97	0.66	0.90
NB	0.91	0.92	0.49	0.73	0.58	0.52	0.82
RF	0.91	0.92	0.50	0.75	0.99	0.48	0.94
XGB	0.93	0.90	0.42	0.77	0.99	0.48	0.92
SVM	0.91	0.97	0.48	0.75	0.84	0.64	0.92
MLP	0.91	0.97	0.50	0.76	0.99	0.70	0.94
FuzzyGNN	0.93	0.94	0.42	0.75	0.95	0.56	0.87

[†] indicates a revised ANFIS version with feature selection.

Table 9 Precision results across datasets

Model	Iris	Cancer	Haberman	Heart disease	Gas concentration	Ozone	Spambase
ANFIS	0.92	0.92 [†]	0.56	0.77	0.65 [†]	0.50 [†]	0.86 [†]
TSK	0.86	0.96	0.50	0.74	0.98	0.75	0.92
SHAP-FuzzyGNN	0.88	0.94	0.36	0.70	0.88	0.71	0.85
DT	0.87	0.92	0.54	0.76	0.97	0.64	0.90
LR	0.84	0.97	0.37	0.75	0.97	0.87	0.91
NB	0.92	0.93	0.52	0.74	0.66	0.56	0.83
RF	0.92	0.93	0.54	0.75	0.99	0.47	0.95
XGB	0.93	0.93	0.37	0.77	0.99	0.47	0.93
SVM	0.92	0.97	0.51	0.75	0.87	0.75	0.92
MLP	0.93	0.97	0.51	0.76	0.99	0.72	0.94
FuzzyGNN	0.93	0.75	0.37	0.75	0.95	0.76	0.88

[†] indicates a revised ANFIS version with feature selection.

Table 10 Recall results across datasets

Model	Iris	Cancer	Haberman	Heart disease	Gas concentration	Ozone	Spambase
ANFIS	0.91	0.92 [†]	0.53	0.77	0.62 [†]	0.47 [†]	0.82 [†]
TSK	0.86	0.94	0.51	0.74	0.98	0.68	0.93
SHAP-FuzzyGNN	0.88	0.94	0.50	0.70	0.87	0.50	0.82
DT	0.87	0.92	0.54	0.76	0.97	0.52	0.89
LR	0.80	0.96	0.50	0.75	0.97	0.61	0.90
NB	0.91	0.92	0.51	0.74	0.57	0.74	0.84
RF	0.91	0.91	0.52	0.75	0.99	0.50	0.94
XGB	0.93	0.89	0.50	0.77	0.99	0.50	0.92
SVM	0.91	0.97	0.50	0.75	0.84	0.60	0.91
MLP	0.91	0.98	0.51	0.76	0.99	0.69	0.94
FuzzyGNN	0.93	0.94	0.50	0.75	0.95	0.54	0.86

[†] indicates a revised ANFIS version with feature selection.

Acknowledgements Gianluca Zaza and Giovanna Castellano acknowledge the support of the PNRR project FAIR—Future AI Research (PE00000013), Spoke 6—Symbiotic AI (CUP H97G22000210007) under the NRRP MUR program funded by the NextGenerationEU. A Ph.D. fellowship funds Raffaele Scaringi’s research within the Italian "D.M. n. 352, April, 9, 2022" under the NRRP, Mission 4, Component 2, Investment 3.3 Ph.D. project "Automatic analysis of artistic heritage via Artificial Intelligence" co-supported by Exprivia S.p.A (CUP H91I22000410007). Gianluca Zaza, Giovanna Castellano and Raffaele Scaringi are INdAM GNCS research group members. We acknowledge the CINECA award under the ISCRA initiative, which gave us access to computing resources and support.

Author contributions Gianluca Zaza: Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Software, Writing—Original draft. Raffaele Scaringi: Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Software, Writing—Original draft. Gennaro Vessio: Conceptualization, Investigation, Supervision, Validation, Writing – Review & editing. Giovanna Castellano: Conceptualization, Investigation, Supervision, Validation, Writing – Review & editing.

Funding Open access funding provided by Università degli Studi di Bari Aldo Moro within the CRUI-CARE Agreement.

Declarations

Conflict of interest The authors declare that they have no known conflict of financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Consent for publication We, the undersigned authors, hereby certify that we have seen and approved the final version of the manuscript titled “FuzzyGNN: A Graph Neural Network Framework for Enhanced Fuzzy Modeling” submitted to Neural

Computing and Applications. We warrant that the article is our original work, has not received prior publication, and is not under consideration for publication elsewhere. Each of us has made substantial contributions to the research study, reviewed the manuscript, and approved its content.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Von Eschenbach WJ (2021) Transparency and the black box problem: why we do not trust AI. *Philos Technol* 34(4):1607–1622
2. Arrieta AB, Díaz-Rodríguez N, Del Ser J, Bennetot A, Tabik S, Barbado A, García S, Gil-López S, Molina D, Benjamins R et al (2020) Explainable artificial intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI. *Inform fusion* 58:82–115
3. Zadeh LA (1965) Fuzzy sets. *Inf Control* 8(3):338–353
4. Jin Y (2000) Fuzzy modeling of high-dimensional systems: complexity reduction and interpretability improvement. *IEEE Trans Fuzzy Syst* 8(2):212–221
5. Kliegr T, Bahník Fúrnkranz J (2021) A review of possible effects of cognitive biases on interpretation of rule-based machine learning models. *Artif Intell* 295:103458
6. Zhou Y, Zheng H, Huang X, Hao S, Li D, Zhao J (2022) Graph neural networks: taxonomy, advances, and trends. *ACM Trans Intell Syst Technol* 13(1):1–54
7. Ying Z, Bourgeois D, You J, Zitnik M, Leskovec J (2019) Gnnexplainer: generating explanations for graph neural networks. *Adv Neural Inform Process syst* 32
8. Castellano G, Scaringi R, Vessio G, Zaza G (2024) Integrating graph neural networks and fuzzy logic to enhance deep learning interpretability. In: *Proceedings of the First Workshop on Explainable Artificial Intelligence for the Medical Domain (EXPLIMED 2024) Co-located with the 27th European Conference on Artificial Intelligence (ECAI 2024). CEUR Workshop Proceedings, vol. - 3831*
9. Ribeiro MT, Singh S, Guestrin C (2016) Model-agnostic interpretability of machine learning. *arXiv preprint arXiv:1606.05386*
10. Lundberg SM, Lee S-I (2017) A unified approach to interpreting model predictions. *Adv Neural Inform Process Syst* 30
11. Alvarez-Melis D, Jaakkola TS (2018) On the robustness of interpretability methods. *arXiv preprint arXiv:1806.08049*
12. Slack D, Hilgard S, Jia E, Singh S, Lakkaraju H (2020) Fooling lime and shap: Adversarial attacks on post hoc explanation methods. In: *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, pp 180–186
13. Ghorbani A, Abid A, Zou J (2019) Interpretation of neural networks is fragile. In: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol 33, pp 3681–3688
14. Scarselli F, Gori M, Tsoi AC, Hagenbuchner M, Monfardini G (2009) The graph neural network model. *IEEE Trans Neural Netw* 20(1):61–80
15. Yuan H, Tang J, Hu X, Ji S (2020) Xggn: towards model-level explanations of graph neural networks. In: *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp 430–438
16. Luo D, Cheng W, Xu D, Yu W, Zong B, Chen H, Zhang X (2020) Parameterized explainer for graph neural network. *Adv Neural Inf Process Syst* 33:19620–19631
17. Maria AJ, Castiello C, Luis M, Mencar C et al (2021) Explainable fuzzy systems: paving the way from interpretable fuzzy systems to explainable AI systems. *Stud Comput Intell* 970:1–253
18. Lee RS (2020) Artificial intelligence in daily life
19. Moezi SA, Zakeri E, Bazargan-Lari Y, Khalghollah M (2014) Fuzzy logic control of a ball on sphere system. *Adv Fuzzy Syst* 2014(1):291430
20. Bazargan-Lari Y, Eghtesad M, Khoogar AR, Mohammad-Zadeh A (2015) Adaptive neural network control of a human swing leg as a double-pendulum considering self-impact joint constraint. *Trans Can Soc Mech Eng* 39(2):201–219
21. Geng C, Zhao Z, Wang Z, Ye H, Si X (2025) Extracting interpretable logic rules from graph neural networks. *arXiv preprint arXiv:2503.19476*
22. Wang Y, Yin T, Zhu S, Ma X, Liang L, Guo J (2025) FIGNN: fuzzy inference-guided graph neural network for fault diagnosis in industrial processes. *IEEE Trans Instrument Measure*

23. Deng Y, Ren Z, Kong Y, Bao F, Dai Q (2017) A Hierarchical fused fuzzy deep neural network for data classification. *IEEE Trans Fuzzy Syst* 25(4):1006–1012
24. Rajurkar S, Verma NK (2017) Developing deep fuzzy network with Takagi Sugeno fuzzy inference system. In: 2017 IEEE Int. Conf. on Fuzzy Systems (FUZZ-IEEE), pp 1–6
25. Zhou S, Chen Q, Wang X (2014) Fuzzy deep belief networks for semi-supervised sentiment classification. *Neurocomputing* 131:312–322
26. Casalino G, Castellano G, Kaymak U, Zaza G (2021) Balancing accuracy and interpretability through neuro-fuzzy models for cardiovascular risk assessment. In 2021 IEEE Symposium Series on Computational Intelligence (SSCI) (pp. 1–8). IEEE
27. Chimatapu R, Hagrass H, Starkey A, Owusu G (2018) Interval Type-2 Fuzzy Logic Based Stacked Autoencoder Deep Neural Network For Generating Explainable AI Models in Workforce Optimization. In: 2018 IEEE Int. Conf. on Fuzzy Systems (FUZZ-IEEE), pp 1–8
28. Kaczmarek-Majer K, Casalino G, Castellano G, Dominiak M, Hryniewicz O, Kamińska O, Vessio G, Diaz-Rodriguez N (2022) PLENARY: Explaining black-box models in natural language through fuzzy linguistic summaries. *Inf Sci* 614:374–399
29. Fisher RA (1988) Iris. UCI machine learning repository. <https://doi.org/10.24432/C56C76>
30. Wolberg e.a. W (1995) Breast Cancer Wisconsin (Diagnostic). UCI machine learning repository
31. Haberman S (1999) Haberman’s Survival. UCI machine learning repository. <https://doi.org/10.24432/C5XK51>
32. Alizadehsani R, Roshanzamir M, Abdar M, Beykikhoshk A, Khosravi A, Panahiazar M, Koohestani A, Khozeimeh F, Nahavandi S, Sarrafzadegan N, Sarrafzadegan N (2019) A database for using machine learning and data mining techniques for coronary artery disease diagnosis. *Sci Data* 6(1):227
33. Vergara A (2012) Gas sensor array drift at different concentrations. UCI machine learning repository. <https://doi.org/10.24432/C5MK6M>
34. Zhang e.a. (2008) Kun: Ozone Level Detection. UCI machine learning repository. <https://doi.org/10.24432/C5NG6W>
35. Hopkins e.a. (1999) Mark: Spambase. UCI machine learning repository. <https://doi.org/10.24432/C53G6X>
36. Shihabudheen KV, Pillai GN (2018) Recent advances in neuro-fuzzy system: a survey. *Knowl-Based Syst* 152:136–162
37. Guo F, Liu J, Li M, Huang T, Zhang Y, Li D, Zhou H (2022) A concise TSK fuzzy ensemble classifier integrating dropout and bagging for high-dimensional problems. *IEEE Trans Fuzzy Syst* 30(8):3176–3190
38. Wang H, Liang Q, Hancock JT, Khoshgoftaar TM (2024) Feature selection strategies: a comparative analysis of SHAP-value and importance-based methods. *J Big Data* 11(1):44
39. Safavian SR, Landgrebe D (1991) A survey of decision tree classifier methodology. *IEEE Trans Syst Man Cybern* 21(3):660–674
40. Helmreich JE (2016) Regression modeling strategies with applications to linear models, logistic and ordinal regression and survival analysis (2 edition). *J Stat Softw* 70:1–3
41. Yang F-J (2018) An implementation of naive bayes classifier. In: 2018 International Conference on Computational Science and Computational Intelligence (CSCI), pp 301–306
42. Breiman L (2001) Random Forests. *Mach Learn* 45:5–32
43. Chen T, Guestrin C (2016) XGBoost: a scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining
44. Hearst MA, Dumais ST, Osuna E, Platt J, Scholkopf B (1998) Support vector machines. *IEEE Intell Syst Appl* 13(4):18–28
45. Rumelhart DE, McClelland JL (1987) Learning Internal Representations by Error Propagation, pp 318–362
46. Guillaume S (2001) Designing fuzzy inference systems from data: an interpretability-oriented review. *IEEE Trans Fuzzy Syst* 9(3):426–443
47. Brody S, Alon U, Yahav E (2021) How attentive are graph attention networks? arXiv preprint [arXiv:2105.14491](https://arxiv.org/abs/2105.14491)
48. Bresson X, Laurent T (2017) Residual gated graph convnets. arXiv preprint [arXiv:1711.07553](https://arxiv.org/abs/1711.07553)
49. Kipf TN, Welling M (2016) Semi-supervised classification with graph convolutional networks. arXiv preprint [arXiv:1609.02907](https://arxiv.org/abs/1609.02907)
50. Loshchilov I, Hutter F (2017) Decoupled weight decay regularization. arXiv preprint [arXiv:1711.05101](https://arxiv.org/abs/1711.05101)
51. Wu Z, Pan S, Chen F, Long G, Zhang C, Yu PS (2020) A comprehensive survey on graph neural networks. *IEEE Trans Neural Netw Learn Syst* 32(1):4–24

Publisher’s Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Gianluca Zaza¹ · Raffaele Scaringi¹ · Gennaro Vessio¹ · Giovanna Castellano¹

✉ Gianluca Zaza
gianluca.zaza@uniba.it
Raffaele Scaringi
raffaele.scaringi@uniba.it
Gennaro Vessio
gennaro.vessio@uniba.it
Giovanna Castellano
giovanna.castellano@uniba.it

¹ Department of Computer Science, University of Bari Aldo Moro, Bari, Italy